MACHINE LEARNING

Reinforcement Learning Visual Guide

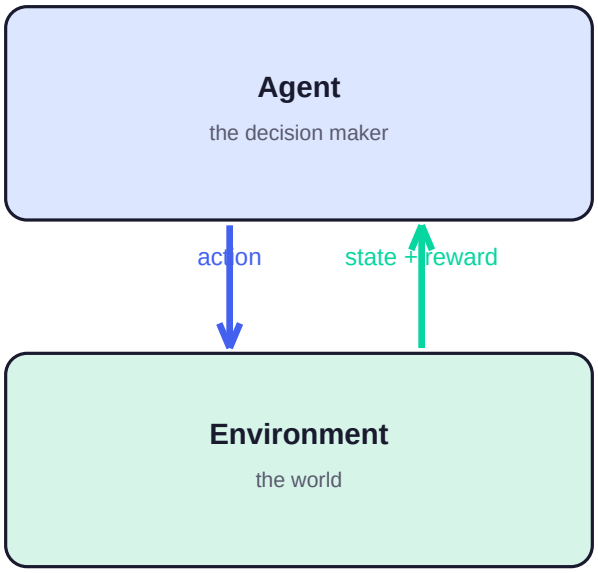
Learning by doing:
agents, rewards, Q-learning, policy gradients.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

The Agent-Environment Loop

Act, observe, get rewarded, repeat.



At each step the agent sees a state, takes an action, and receives a reward. Goal: maximize total reward.

No labeled data — the reward signal IS the supervision. Sparse or delayed rewards make RL hard.

STEP 2/6

Rewards and Returns

Not just now — the whole future, discounted.

Reward r(t)

immediate

Return G(t)

sum of future rewards, discounted.

Discount

future rewards worth less.

gamma < 1.

G(t) = r(t) + gamma*r(t+1) + gamma^2*r(t+2) + ... — discounting prefers sooner rewards and bounds infinity.

Reward shaping (small hints along the way) speeds learning — but a misshaped reward teaches the wrong behavior.

STEP 3/6

Exploration vs Exploitation

Try new things, or milk what works?

Explore

discover better strategies.

random actions:

Exploit

collect reliable reward.

best known action:

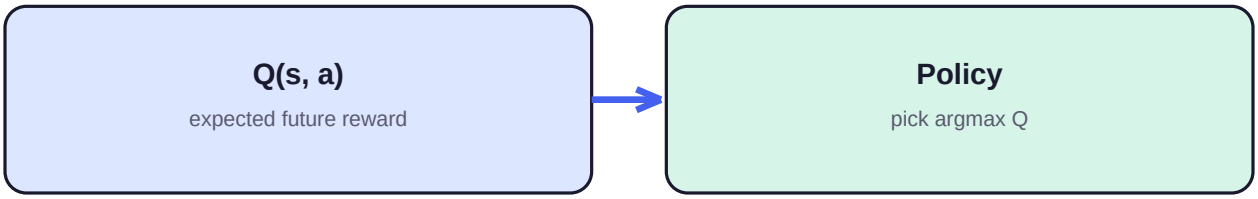
Epsilon-greedy: act randomly with probability epsilon, greedily otherwise. Decay epsilon over training.

Too much exploration never converges; too little gets stuck in a local optimum. The central tension of RL.

STEP 4/6

Q-Learning

Learn the value of every action in every state.



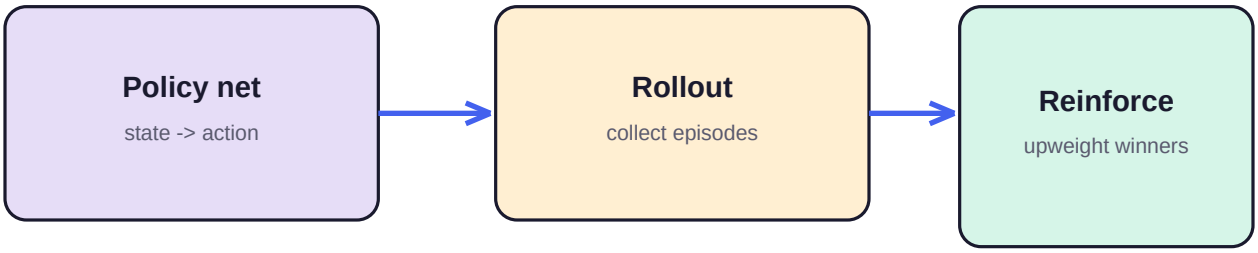
Update: Q(s,a) += alpha * [r + gamma*max Q(s',a') - Q(s,a)]. Bootstrap from your own estimates.

Deep Q-Networks replace the table with a neural net. Use a frozen target network and experience replay for stability.

STEP 5/6

Policy Gradients

Optimize the policy directly.



Increase the probability of actions that led to high reward. REINFORCE, PPO, and actor-critic refine this idea.

PPO (clipped updates) is the workhorse — including inside RLHF for language models.

STEP 6/6

RL in Practice

Where it shines — and what to watch.

- Games and robotics: simulators give unlimited cheap episodes — the ideal RL setting.
- RLHF: human preference becomes the reward; PPO/DPO aligns language models.
- Sample inefficiency is the tax: millions of episodes for what a human learns in minutes.
- Always try supervised or imitation learning first — reach for RL when no labels exist.
- Evaluate on held-out environments: RL agents overfit to their simulators.

STEP ✓/✓

Key Takeaways

Reinforcement Learning — Remember

- Agent acts, environment returns state + reward; maximize discounted return.
- Discount factor gamma trades off immediate vs future reward.
- Epsilon-greedy balances exploration (discover) vs exploitation (harvest).
- Q-learning values state-action pairs; policy gradients optimize the policy directly (PPO).
- RL needs simulators or cheap episodes; prefer imitation learning when labels exist.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev